

Compositional Non-Face Re-Identification Pressure under Cumulative Vision Releases

Auditing the “leaky bucket” effect in vision benchmarks

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Main message. Removing faces does not make identity risk “reset” with each new release. Dataset expansions, checkpoints, embeddings, indices, and metadata accumulate as side information. We define $vRPI_\alpha$, prove that true-posterior pressure is non-decreasing under added releases, and show large empirical pressure growth on face-masked Re-ID benchmarks.

1. Why Single-Release Privacy Is the Wrong Unit

Vision privacy evaluations usually inspect one release in isolation. Real ecosystems publish artifacts sequentially: data, checkpoints, embeddings, indices, sometimes metadata.

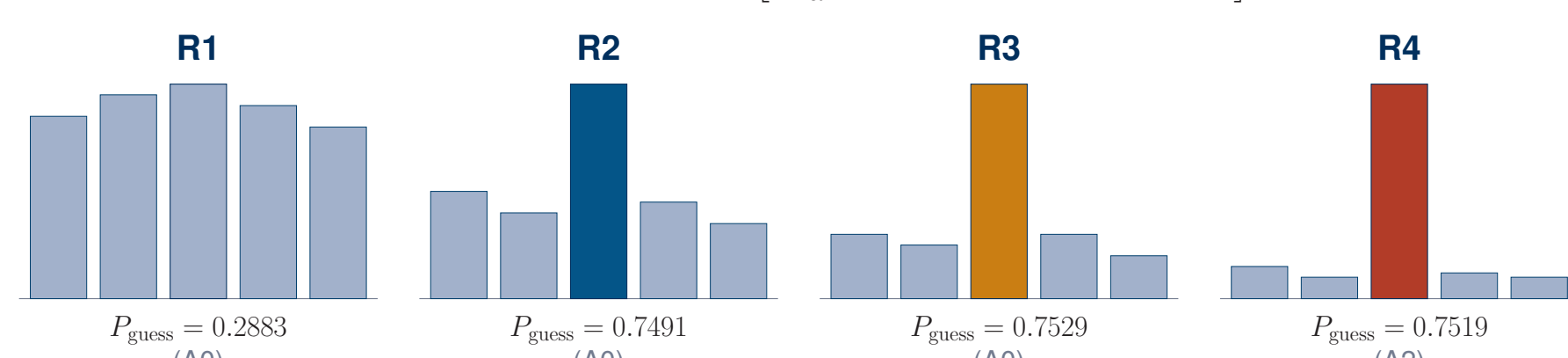
Latent identity. Let $U \in \{1, \dots, N\}$ index identity and X be a raw observation. The non-face transform

$$\bar{X} = \tau(X)$$

removes faces, but identity remains in clothing, gait, and scene context.

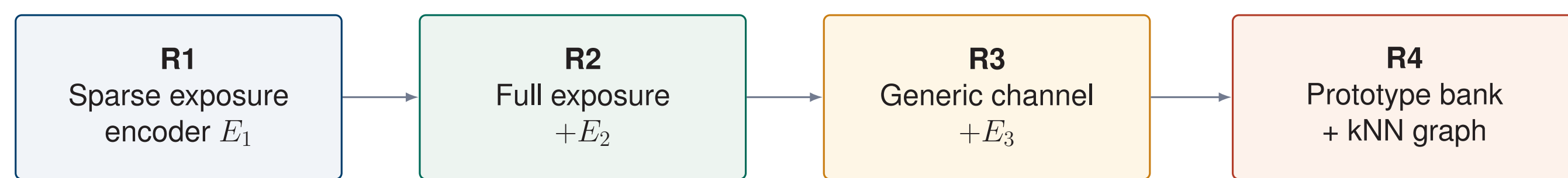
The compositional question. Does the posterior $\Pr(U \mid Z_{1:T})$ sharpen as releases accumulate? The operational target is Bayes-optimal guessing:

$$P_{\text{guess}}(U \mid Z_{1:T}) = \mathbb{E}[\max_u \Pr(U = u \mid Z_{1:T})].$$



Sketch of $\Pr(U \mid Z_{1:t})$ concentrating across releases – bars are illustrative, not measured. Measured P_{guess} is printed below each panel (Table, §4).

2. Release-as-Channel Formulation



$Z_{1:T} = (Z_1, \dots, Z_T)$ accumulates side information

$\Pr(U \mid Z_{1:T})$ becomes the object of the privacy audit

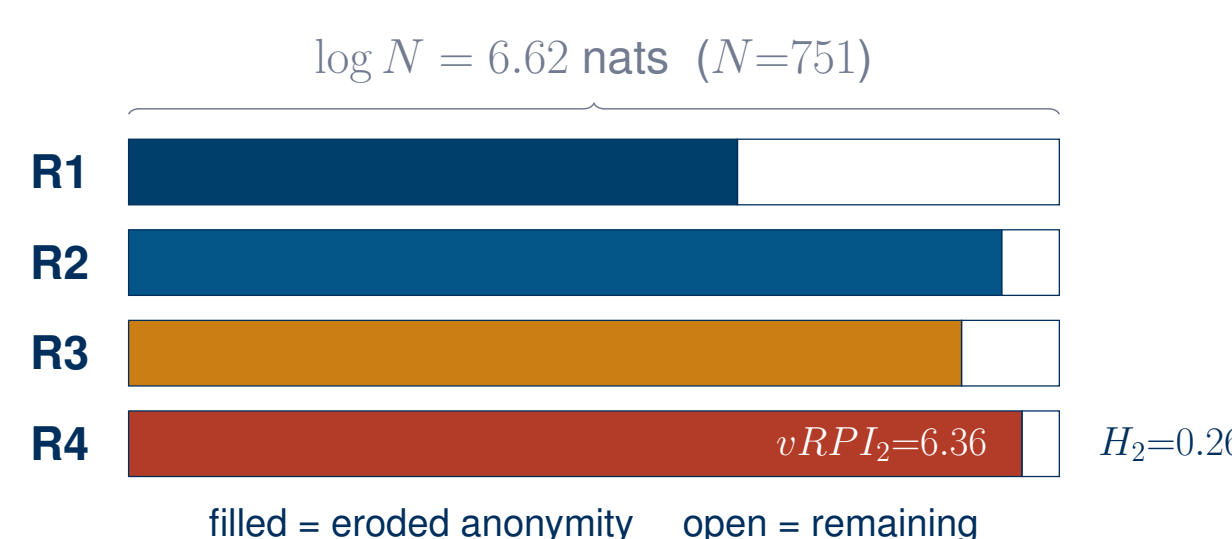
Each disclosure is modeled as a channel $Z_t \sim K_t(\cdot \mid \bar{X})$. The framework is release-sequence aware rather than tied to one Re-ID model or anonymizer.

3. $vRPI_\alpha$: Metric and Core Theory

Rényi pressure index. For $\alpha > 1$, using Arimoto conditional Rényi entropy,

$$vRPI_\alpha(T) = \log N - H_\alpha(U \mid Z_{1:T}).$$

Under a uniform prior, $\log N$ is maximal identity uncertainty: $vRPI_\alpha$ is the anonymity already eroded by cumulative releases.



Rows R1–R4: filled = eroded $vRPI_2$ (R4 uses attacker A2, others A0); open = remaining $H_2(U \mid Z_{1:T})$. Nats, $N = 751$.

Theorem 1: monotonicity on the true posterior

$$H_\alpha(U \mid Z, W) \leq H_\alpha(U \mid Z) \implies vRPI_\alpha(T+1) \geq vRPI_\alpha(T).$$

Added side information cannot restore identity uncertainty.

Theorem 2: operational guessing bridge

$$P_{\text{guess}}(U \mid Z) \leq \exp\left(\frac{1-\alpha}{\alpha} H_\alpha(U \mid Z)\right).$$

Thus higher pressure corresponds to a smaller effective anonymity set and stronger identity inference.

Choosing α . $\alpha = 2$ is the recommended stable default; $\alpha = 5$ or 10 emphasizes high-confidence / worst-case posterior concentration and should be paired with calibration diagnostics.

Additional guarantees.

- **Ideal additivity benchmark:** under conditional independence given identity, Sibson information factorizes into an additive upper bound on pressure across releases.
- **Ablation monotonicity:** post-processing that removes information cannot increase measured pressure.
- **Plug-in stability:** pressure estimated from an approximate posterior changes continuously, with an explicit logarithmic error bound.

10. Final Contribution, Empirical Signal, and Limitation

Contribution

A **release-sequence privacy formalism**, $vRPI_\alpha$, from Arimoto conditional Rényi entropy.

Theory: monotonicity under added side information, a Bayes-guessing bridge, an additive benchmark, ablation monotonicity, and plug-in stability.

Empirical signal

On face-masked Market-1501, **exposure expansion dominates:** A0 P_{guess} .288→.749, $vRPI_2$ 4.332→6.212.

CUHK03 preserves the paired R1→R2 direction; prototype/index and kNN-graph releases add artifact-specific leakage.

Limitation

Evaluation is theory-aligned and closed-world. $vRPI_\alpha$ is an **audit statistic**, not a mitigation.

Next: **open-set posteriors**, **realistic anonymizers**, **more datasets/video**, and **longer real release sequences**.

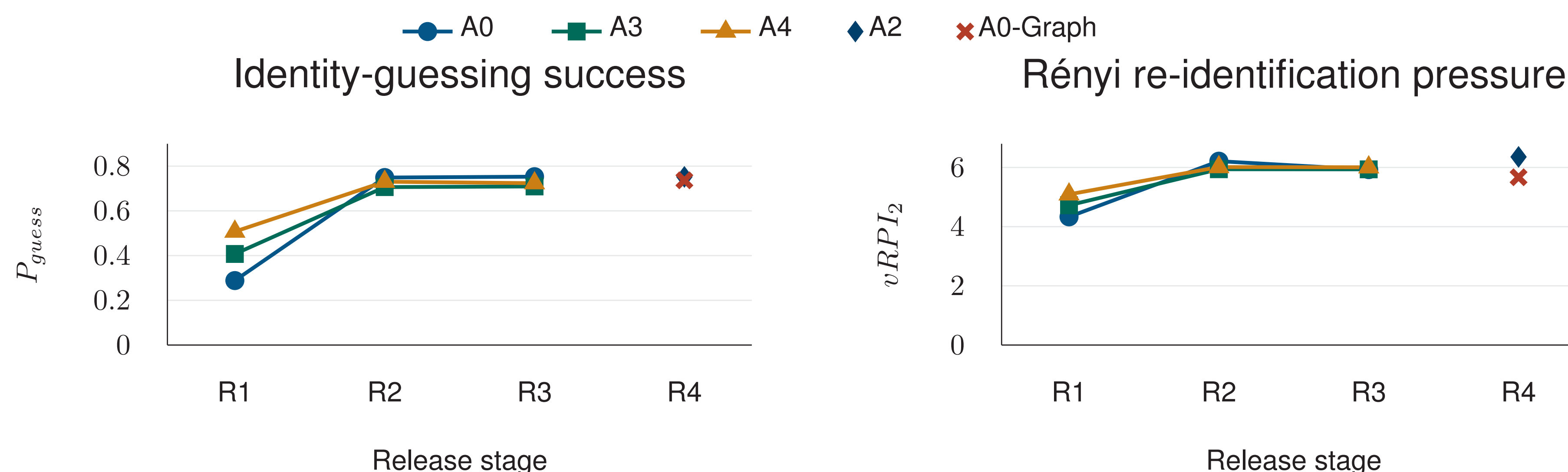
4. Main Empirical Result on Market-1501- τ

The dominant leak is exposure expansion. From R1 to R2, retrieval attacker A0 jumps from $P_{\text{guess}} = 0.288$ to 0.749 and $vRPI_2 = 4.332$ to 6.212. A3 rises from 0.407 to 0.706. R4 then shows that publishable artifacts themselves create exploitable leakage pathways.

Release	Attacker	P_{guess}	$vRPI_2$	Interpretation
R1	A0	.2883	4.3319	Sparse exposure baseline
R2	A0	.7491	6.2119	Dominant exposure-expansion jump
R3	A0	.7529	5.9271	Generic channel; calibrated fusion
R4	A2	.7519	6.3572	Prototype/index specialist
R4	A0-Graph	.7366	5.6697	kNN-graph pathway

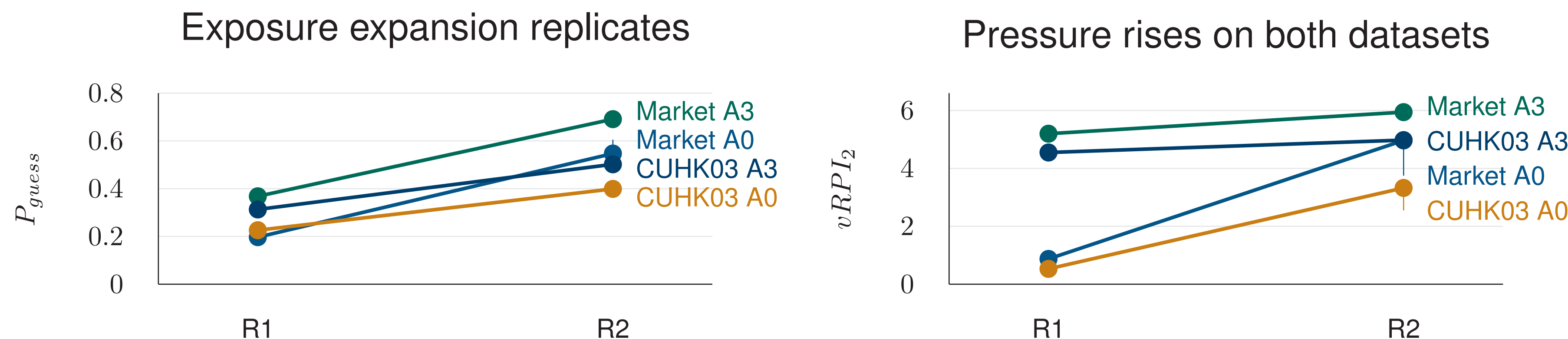
Mean over 3 seeds. A3 independently shows the same R1→R2 pattern (P_{guess} : .407→.706; $vRPI_2$: 4.723→5.940). Empirical plug-in estimates can fluctuate with attacker/calibration even though Theorem 1 is a true-posterior statement.

5. Release Trend: The R1→R2 Jump Dominates



The exposure expansion is the clearest compositional effect. R2→R3 is stable under validation-tuned fusion and temperature scaling. R4 demonstrates artifact-specific leakage through the prototype/index specialist and graph-exploiting attacker.

6. Replication + Non-Face Stress Test



CUHK03- τ replication. A lighter paired protocol still yields a positive R1→R2 increase for both retrieval (A0) and nonlinear-probe (A3) attackers on Market and CUHK03. The absolute magnitudes are not compared across protocols; the paired direction is the claim.

Masking ablation. Sweeping top-region masking from 0% to 40% leaves pressure essentially flat: A0 $vRPI_2 \in [4.87, 4.91]$, A3 $vRPI_2 \in [5.83, 5.88]$. Even at 40% masking, $P_{\text{guess}} \geq .51$ (A0) and $\geq .64$ (A3).

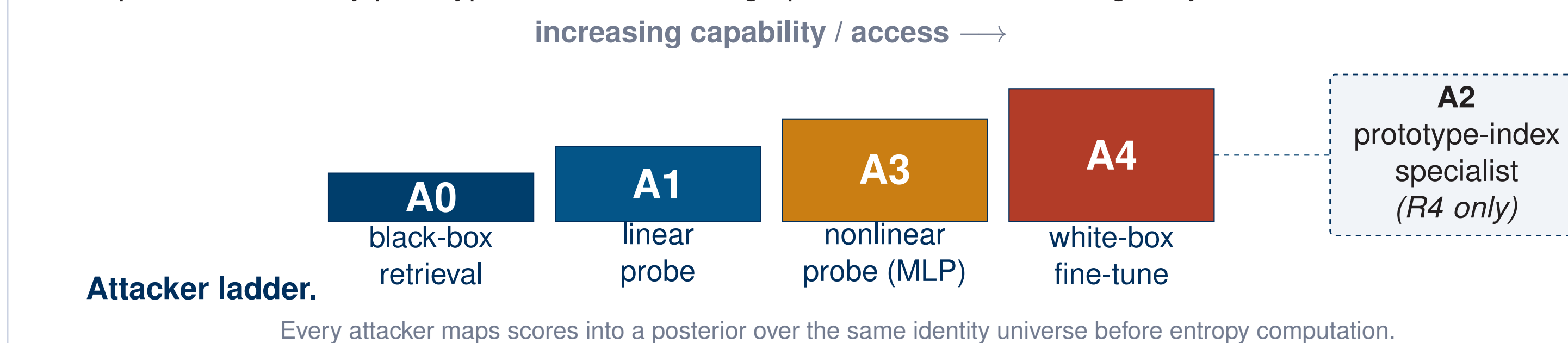
Important scope. The top-region mask is a controlled proxy, not a realistic detector-based or generative anonymizer. The result shows persistent non-face signal under aggressive region removal; it does not claim that all anonymization methods fail.

7. Experimental Protocol and Attacker Ladder

Dataset. Main experiment: Market-1501 training identities, $N = 751$, with the top 20% of each image masked (Market-1501- τ). Images are split into exposure, gallery, query, and held-out validation pools over the same closed-world label space.

What counts as a release?

- **R1:** encoder E_1 trained on sparse exposure.
- **R2:** add E_2 trained on full exposure; cumulative feature state.
- **R3:** add generic ImageNet-pretrained ResNet-50 E_3 .
- **R4:** publish an identity prototype bank and a kNN graph derived from the R3 gallery.



Every attacker maps scores into a posterior over the same identity universe before entropy computation.

8. Diagnostics: Dependence, Calibration, Priors

Dependence overlap. Conditional independence gives an ideal additive benchmark, not a claim about real CV releases. On Market,

$$\Delta_{\text{cov},2} = \sum_i \widehat{vRPI}_2(Z_i) - \widehat{vRPI}_2(Z_{1:T}) = 0.304 > 0,$$

quantifying redundancy among correlated release channels.

Temperature calibration. Across $\theta \in \{.025, .05, .1, .2, .5, 1.0\}$, absolute pressure shifts but $\Delta vRPI_2(R2 - R1)$ stays positive throughout: A0 spans $[-.781, 4.100]$, A3 spans $[-.567, .677]$.

Non-uniform identity priors. For a Zipf(s) prior, normalizing by prior entropy instead of $\log N$, the R1→R2 increment shrinks from 3.818 at $s = 0$ to .123 at $s = 1.5$: a concentrated prior leaves less anonymity to erode.

9. What the Framework Adds

Not another Re-ID model. $vRPI_\alpha$ audits release-induced privacy pressure; rank/mAP remains a retrieval-utility metric – different questions.

Not a replacement for maximal leakage or DP. Maximal leakage optimizes adversarial gain per channel; DP gives worst-case stability guarantees. $vRPI_\alpha$ fixes the identity task and measures posterior concentration over a release sequence.

Governance implication. Privacy review should happen at the **sequence level**: evaluate planned disclosures together, minimize unnecessary metadata, and treat embedding dumps, prototype banks, and indices as privacy-relevant releases.

Positioning against common criteria.

Measure	Object measured	Sequence-aware?
Rank/mAP	Retrieval ranking quality	No
P_{guess}	Bayes posterior maximum	Yes, on $Z_{1:T}$
$vRPI_\alpha$	Rényi posterior concentration	Yes
Maximal leakage	Worst-case channel gain	Channel-level
DP / attribute inference	Worst-case stability / bound	Via composition

Release hygiene, not surveillance. Intended use is governance: audit successive disclosures before publication, limit high-leverage artifacts, minimize unnecessary metadata.

Limitations.

- Closed-world image benchmarks; broader datasets, video, larger scale, and open-set evaluation remain future work.
- Top-region masking is a controlled proxy, not realistic detector-based or generative anonymization.
- $vRPI_\alpha$ is an audit statistic, not a mitigation method.